Software Architecture and Design Specification

Corporate Mental Wellness & Counseling Platform (Problem Statement #14)

Name: Dhanush Shankar Reddy
SRN: PES1UG24AM083
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Revision history

Version	Date	Author	Change summary
1.0	28-10-2026	Dhanush Shankar Reddy	First version of the architecture and design specification

Approvals

Role	Name	Signature/Date
Course Coordinator		

1. Introduction

1.1 Purpose

This document specifies the architecture and design of the Corporate Mental Wellness and Counseling Platform: how it is split into services, why we split it that way, and how the services talk to each other.

1.2 Scope

It covers the four backend services, the API gateway, the data stores, the main runtime flows, the public interfaces and the security design.

1.3 Audience

Developers, QA engineers, and the course evaluators.

1.4 Definitions

API gateway: the single entry point from the internet. Pseudo-ID: the random identifier used in place of a name. IMappingStore, IPseudonym, INotify: the named interfaces between components. AES-256: the encryption on the identity mapping. STRIDE: the threat categories used in section 3.9.

2. Document overview

2.1 How to use this document

Section 3 is the architecture: the components, the pattern and the reasons for it, the risks, and the security design. Section 4 is the design: the runtime flows as sequence diagrams, the API definitions, and how errors and logging are handled.

2.2 Related documents

SRS, STP, and the RTM in the SRS.

3. Architecture

3.1 Goals & constraints

Goals: keep the identity of every user hidden, keep working when one part fails, and load the slot list in under 2 seconds at peak times. Constraints: the identity mapping is stored separately and encrypted with AES-256, and no report is produced for a group of fewer than ten people.

3.2 Stakeholders & concerns

- Employees: real anonymity and a fast, clear portal.
- Therapists: availability management and session notes by pseudo-ID.
- Administrator: aggregate trends without ever seeing individuals.
- Developers: parts that can be built, deployed and scaled on their own.

3.3 Component (UML) diagram

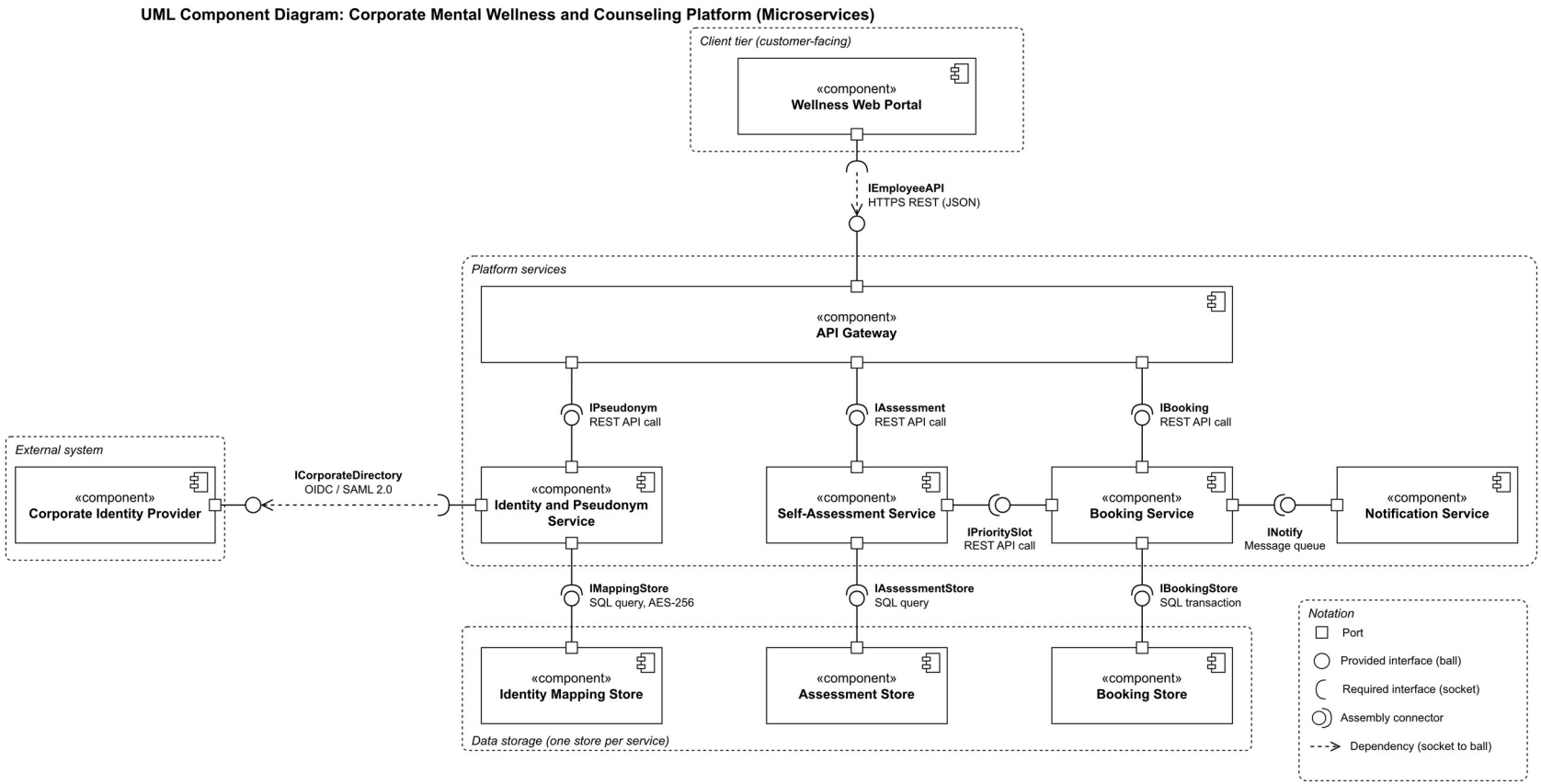


Figure 1. UML component diagram

3.4 Component descriptions

- Wellness Web Portal: the browser front end; talks only to the gateway.
- API Gateway: single entry point; checks the token through IPseudonym and forwards only the pseudo-ID.
- Identity and Pseudonym Service: the one component that can read the identity mapping, through IMappingStore; backed by the encrypted Identity Mapping Store.
- Self-Assessment Service: scores the questionnaire, raises the crisis path, and builds the trend view and the aggregate report; backed by the Assessment Store.
- Booking Service: manages therapist availability, lists and reserves slots, and holds session notes; backed by the Booking Store.
- Notification Service: sends confirmations, reached through INotify over a message queue.

3.5 Chosen architecture pattern and rationale

Microservices. The platform is split into the four services above behind one gateway, each deployed on its own, each data-holding service with its own store. A layered monolith was rejected because it would keep the identity mapping in the same process as assessment and booking data, so anonymity would depend on every developer following a rule rather than on the deployment itself.

The split isolates the identity mapping in one service, so NFR-002 is enforced by deployment. It lets the Booking Service scale on its own for peak load without scaling the rest. It also keeps a failure inside one service, so the assessment still works and still shows the crisis helpline when the Notification Service is down.

3.6 Technology stack & data stores

A web portal over HTTPS, backend services exposing JSON APIs, a relational store per data-holding service, and a message queue between the Booking and Notification services. The Identity Mapping Store is encrypted with AES-256.

3.7 Risks & mitigations

Risk	Mitigation
A breach of the booking or assessment store exposes users.	Those stores hold only pseudo-IDs, scores and slots; names live only in the encrypted mapping.
The slot list slows down at peak booking times.	Scale the Booking Service alone; its queries run against its own store.
The Notification Service is down when a booking is made.	Confirmations go through a queue, so the booking still completes.

3.8 Traceability to requirements

- FR-001, FR-006, FR-007, NFR-002, SR-002 → Identity and Pseudonym Service.
- SR-003, SR-004 → API Gateway.
- FR-002, FR-004, FR-005, FR-008, FR-009, FR-015 → Self-Assessment Service.

- FR-003, FR-010, FR-012 to FR-014, NFR-001 → Booking Service.
- FR-011 → Notification Service through INotify.

3.9 Security architecture

Threat modeling (STRIDE): Spoofing is handled by the token check at the gateway. Tampering and information disclosure are handled by TLS in transit and AES-256 on the identity mapping at rest. A denial-of-service burst is absorbed by scaling the Booking Service on its own. Elevation of privilege is limited because only the Identity service can read the mapping, through IMappingStore, and every other service receives a pseudo-ID and nothing more.

4. Design

4.1 Design overview

Each service owns its data and exposes a small JSON interface. The portal drives everything through the gateway. The two flows that matter most are the self-assessment with its crisis path, and the booking of a session; both are drawn as sequence diagrams below.

4.2 UML sequence diagrams

Flow 1: an employee completes the self-assessment, the Self-Assessment Service scores it, and on a severe score it shows the crisis helpline and offers a priority slot. Flow 2: an employee picks a slot, the Booking Service re-checks and reserves it in one transaction, then hands the confirmation to the Notification Service through the queue.

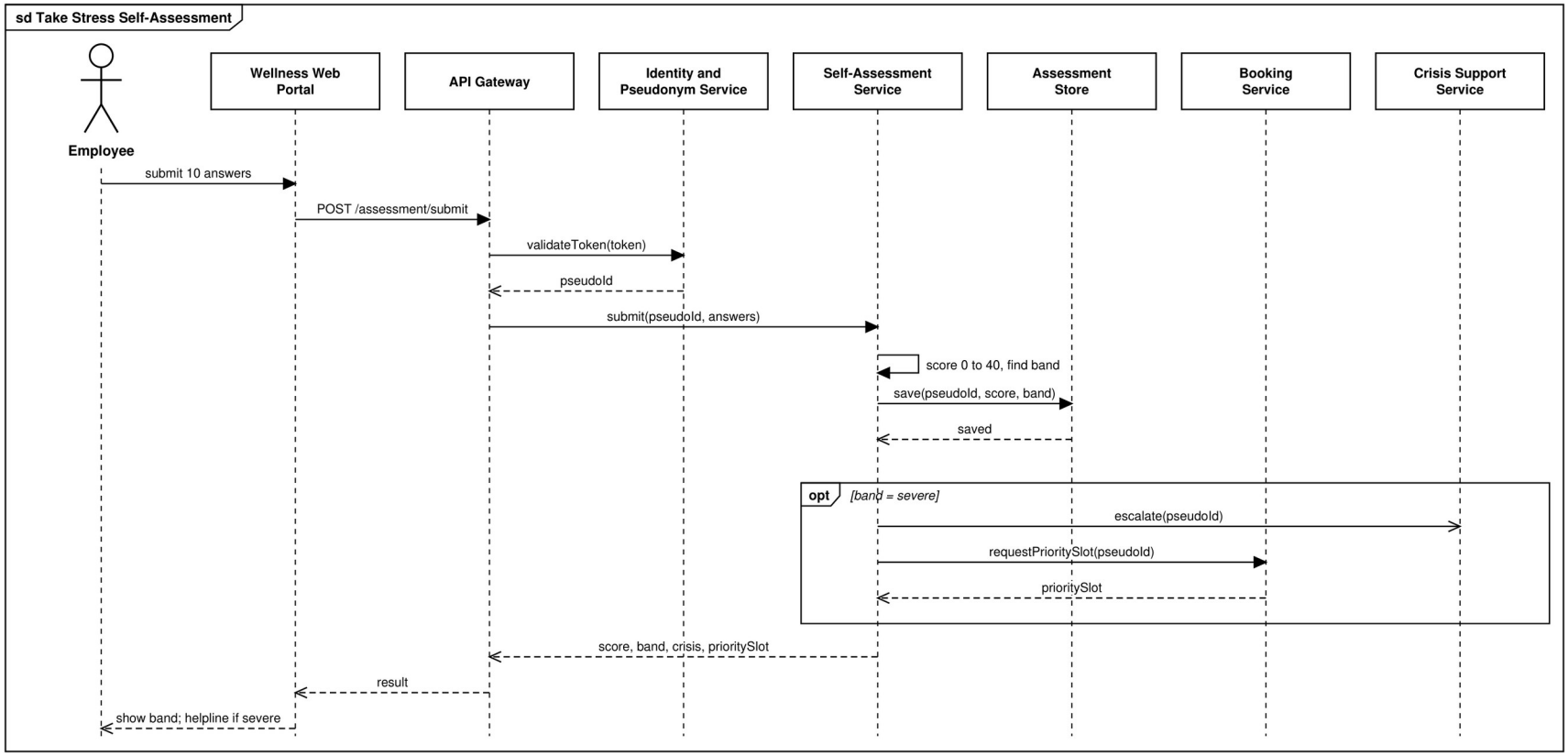


Figure 2. Sequence diagram 1: Take Stress Self-Assessment, with crisis escalation

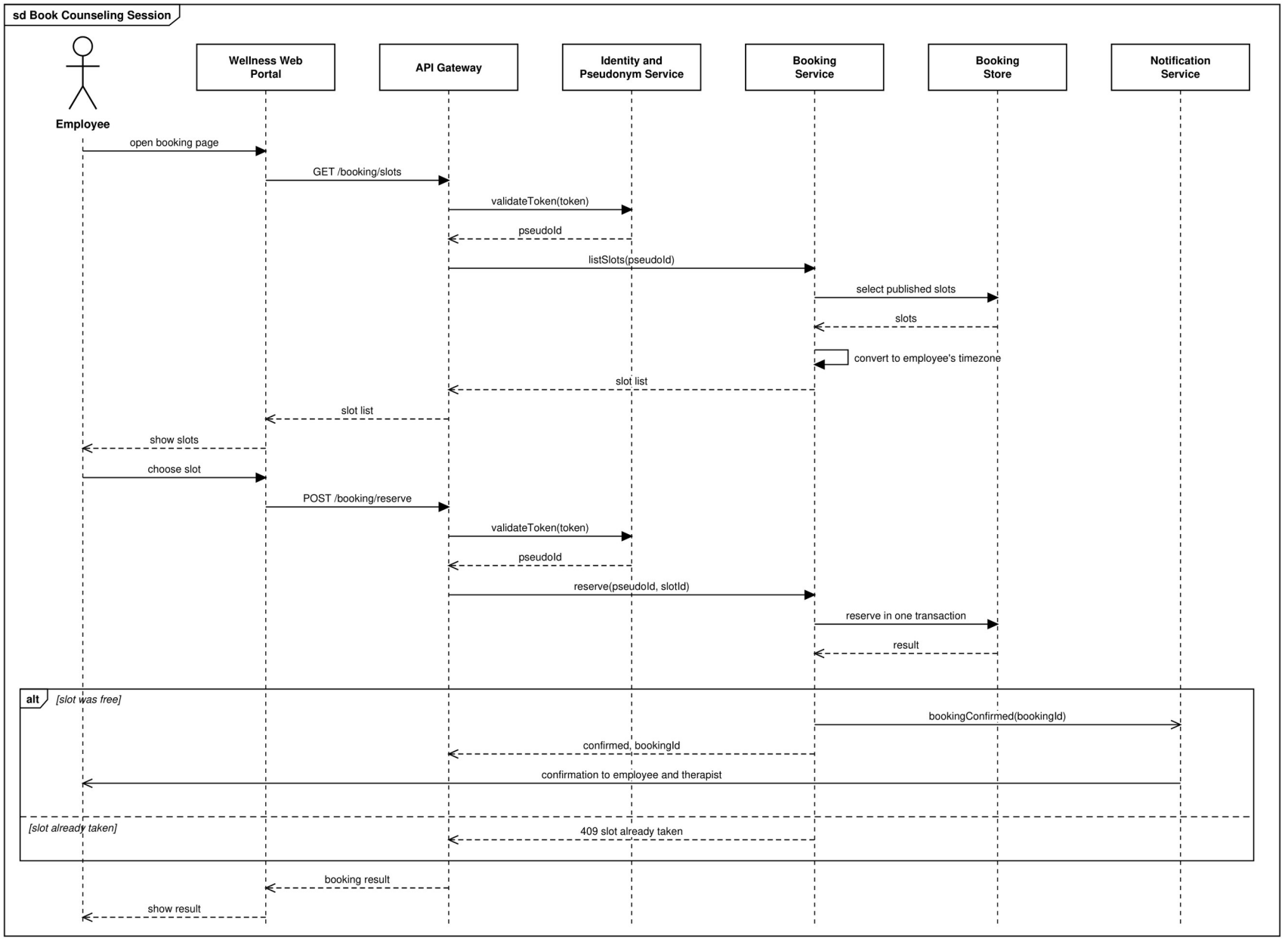


Figure 3. Sequence diagram 2: Book Counseling Session

4.3 API design

Self-Assessment Service

Endpoint: /assessment/submit
Method: POST
Request: {pseudoId, answers[10]}
Response: {score, band, crisis}
Errors: 400 Invalid answer set, 401 Invalid token.

Booking Service

Endpoint: /booking/reserve
Method: POST
Request: {pseudoId, slotId}
Response: {status, bookingId}
Errors: 401 Invalid token, 409 Slot already taken.

4.4 Error handling, logging & monitoring

Error messages to the employee stay plain and never include anything that could identify them. Logs are keyed by pseudo-ID, never by name, and never record answers to the questionnaire. Monitoring watches slot-list response time, booking success rate, and the health of each service.

4.5 UX design

The assessment is short and reads as a wellness check, not a diagnosis. The crisis helpline appears clearly the moment a severe score is returned. The trend view shows only the employee's own history.

4.6 Open issues & next steps

Future enhancements: session delivery, therapist licensing checks, and billing.

5. Appendices

5.1 Glossary: pseudo-ID, band, API gateway, IMappingStore, IPseudonym, INotify, AES-256, STRIDE, TLS.

5.2 References: IEEE 42010, OWASP, NIST SP 800-160.

5.3 Tools: draw.io.